| DS | **Session** | Searching | **Question Information** | Level 1 Challenge 7 |
|---|---|---|---|---|

Question Description:

Suresh have "N" rectangles.

A rectangle is Silver if the ratio of its sides is in between [1.6, 1.7], both inclusive. Your task is to find the number of silver rectangles.

Constraints:

1 <= N <= 10^5

1 <= W,

H <= 10^9

Input Format:

First line: Integer "N" denoting the number of rectangles Each of the ''N" following lines:

Two integers W, H denoting the width and height of a rectangle

Output Format:

Print the output in a single line contains find the number of Silver rectangles.

Sample Input:

5

10 1

165 100

180 100

170 100

160 100

Sample Output:

3

Explanation:

There are three Silver rectangles: (165, 100), (170, 100), (160, 100).

Logical Test Cases

**Test Case 1**

INPUT (STDIN)

```
4
185 100
180 100
170 100
160 100
```

EXPECTED OUTPUT

```
2
```

**Test Case 2**

INPUT (STDIN)

```
6
185 100
180 100
170 100
160 100
165 90
164 95
```

EXPECTED OUTPUT

```
2
```

Mandatory Test Cases

**Test Case 1**

**Test Case 2**

KEYWORD

```
if(width/height>=1.6 &&
width/height<=1.7)
```

KEYWORD

```
else if(height/width >=1.6
&& height/width<=1.7)
```

Complexity Test Cases

**Test Case 1**

CYCLOMATIC COMPLEXITY

7

**Test Case 2**

TOKEN COUNT

125

**Test Case 3**

NLOC

25

## Code

solution.c

```c
#include <stdio.h>

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
    int silverCount = 0;
    for (int i = 0; i < n; i++) {
        double width, height;
        scanf("%lf %lf", &width, &height);
        if (width / height >= 1.6 && width / height <= 1.7) {
            silverCount++;
        } else if (height / width >= 1.6 && height / width <= 1.7) {
            silverCount++;
        }
    }
    printf("%d\n", silverCount);
    return 0;
}
```

## Output

